University of Windsor

School of Computer Science
https://cs.uwindsor.ca

Master of Applied Computing
COMP-8347
Internet Applications and Distributed Systems

Project Description

Your task is to create a website using Django and Python in PyCharm, which you will learn about during lectures and labs. **Each group will be assigned a variant.** The details are as follows:

Mandatory Features

- Your website should include all the **key elements** covered in lectures and labs, such as creating and editing models, views, templates, forms, user sessions, cookies, authentication, and more.
- Your project must be developed using the **tools introduced in class** (e.g., Python, Django, and PyCharm).
- You must propose your own variant of the website. **Your project must include all the mandatory features outlined in this document.**
- Based on the group size, each student's contribution must be clearly defined and justified. For example, each student may be responsible **for two forms, two models, two templates, two views, a login page, and similar components**. Further details will be discussed during lectures and labs.
- Your website should provide different interfaces for **registered and unregistered users**. As with many popular websites, certain features should only be available to registered users. It is up to your group to decide which features will be restricted and which will remain publicly accessible.
- Your website should include **login, logout, and password recovery ('forgot password')** features.
- Your website should include a **search bar** that allows users to search using different keywords. In addition, a drop-down list should be provided to support search by predefined options.
- Your website should include a **User History** section. You will use cookies and sessions to track user activity on your site (e.g., number of visits per day). For inspiration, you may refer to this example: https://www.care2.com/
- Your website must provide an option for users to **upload** content, such as a photo, document, or ID.
- Your website should include a **footer section** at the bottom of the page, containing features similar to those shown in the figure below.

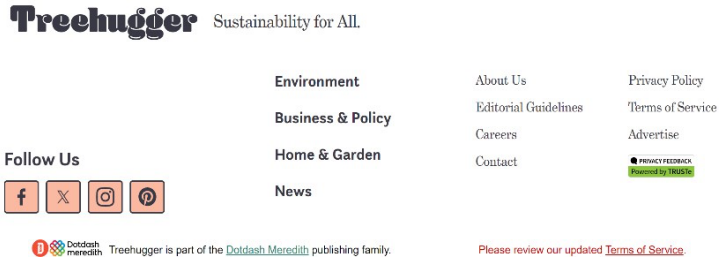

Fig. 1. Sample website

- The website should have a **visually appealing design** and an **eye-catching color scheme**.
- **GitHub** must be used to track the project. Students' progress will be evaluated based on their commits and commit history (including dates) during grading.
- The use of APIs should be kept to a minimum and only employed when no other option is available.

Other Features

- Your database should be saved in **JSON format**, and initial data must be loaded using fixtures.
- Update the index and detail view functions in *views.py* to use class-based views. For reference, see the Django documentation: https://docs.djangoproject.com/en/3.1/topics/class-based-views/).
- Use Bootstrap to style and enhance the appearance of your pages.
- You may also include extra pages, such as Contact Us, About Us, or Team Details.

Groups, Submission, Due Dates, and Marking Criteria

✓ **Group Size:** 5 members per group

✓ **Submission (via Brightspace only):**
Submit a single MS Word document that describes your project. This document must include:
- The steps you followed to create your website.
- The motivation and rationale for choosing your idea (if applicable).
- The role of each team member (preferably shown in a table or chart).
- Related screenshots of your project.
- Links to your live website and the complete code repository on GitHub.

In addition to the Word document, you must also submit the main views.py, models.py, and forms.py files.

Do not submit your files inside a folder. Each required file must be uploaded individually along with the Word document.

✓ **Due Date:** will be posted on Brightspace.
✓ **Total Marks:** 20

You will be evaluated based on the following criteria:

- Individual oral exam/viva (highest weight)
- Your progress and activity on GitHub
- Implementation of mandatory features
- Completion of the project requirements
- Successful execution of your website without errors
- Overall design and visual appeal of your website
- Quality of your presentation
- Inclusion of additional/bonus features
- Quality and completeness of the documents and files submitted on Brightspace

Note 1: As specified in the course outline, the above mark distribution and criteria may be subject to change. For example, based on class performance, all marks may be allocated to the project viva/oral examination and individual contributions.

Note 2: Every group member must contribute to coding using Python/Django/PyCharm, particularly in the views.

Note 3: The use of ChatGPT/other AI tools and/or source code from other external sources will result in a grade of 0.

Note 4: Do not forget to add us to your GitHub repository. Our GitHub ID is ***comp8347proj***.

Presentation Schedule and Details:

- Presentations will take place in **Week 11/12**.
- Each group will have **15 minutes** for the presentation, followed by a **10-minute Q&A session** (subject to change).
- You must prepare a **PowerPoint presentation** *and* provide a **live demo** of your application. Running your application in front of the audience is **mandatory**.
- The presentation order will be determined randomly during or before the class.
- All group members must be **physically present** during the presentation.
- Attendance at all other presentations is **mandatory**.
- **Bonus marks** may be awarded for groups presenting early (Week 11 only).